MTH9877 — Interest Rates & Credit

Assignment 3 — Part E: Extensions

Mortgage Prepayment & Default Survival Models

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E(i) — Competing Risks Framework

(a) Exploratory Data Analysis — Full 34M-Loan Dataset

The Freddie Mac dataset contains 34 million 30-year fixed-rate mortgages originated 1999–2023. Loans exit by prepayment (~57%), remain active / censored (~41%), or default (~2%).

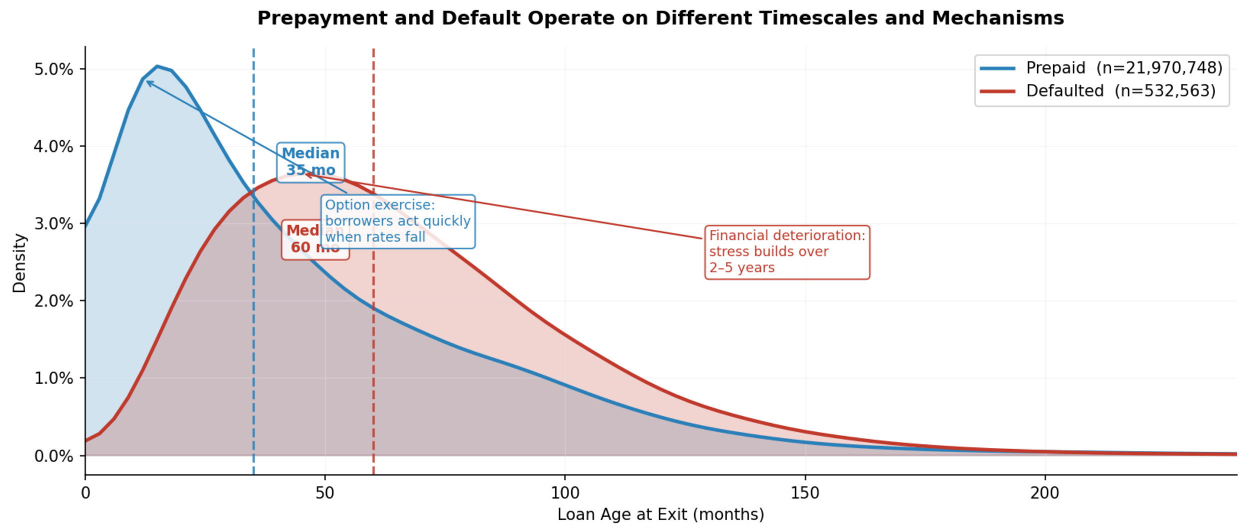

Fig 1. Duration distributions by event type. Prepaid loans exit at a median ~40 months; defaults cluster 30–80 months.

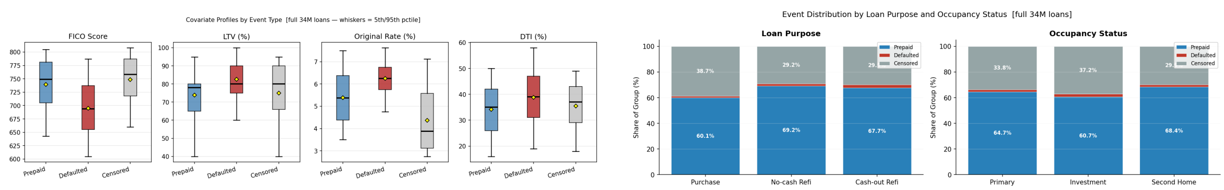

Fig 2. Covariate profiles (whiskers = 5th/95th pctile) and event breakdown by loan purpose and occupancy status.

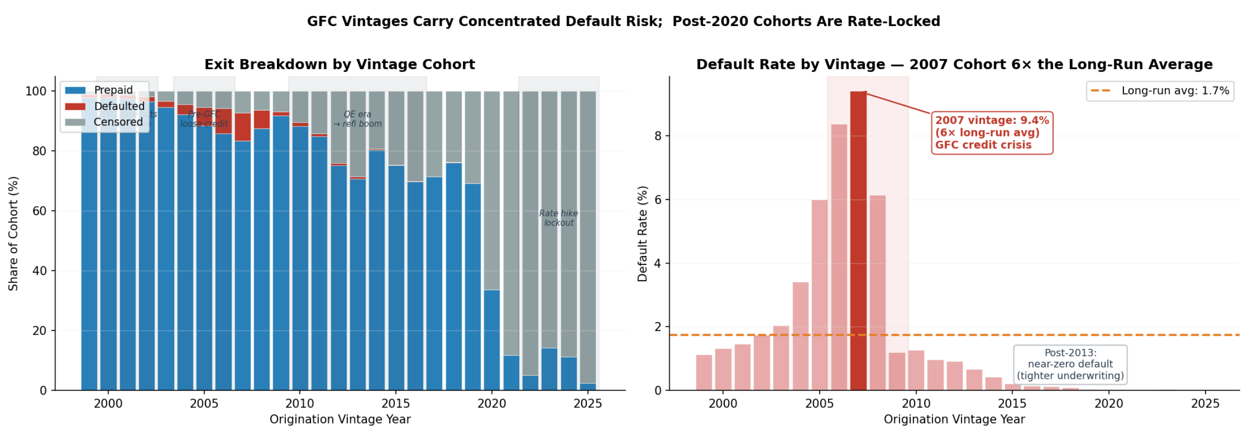

Fig 3. Vintage cohort analysis. 2006–2008 vintages show peak default rates of 8–12%; post-2012 cohorts exhibit near-zero default with high prepayment.

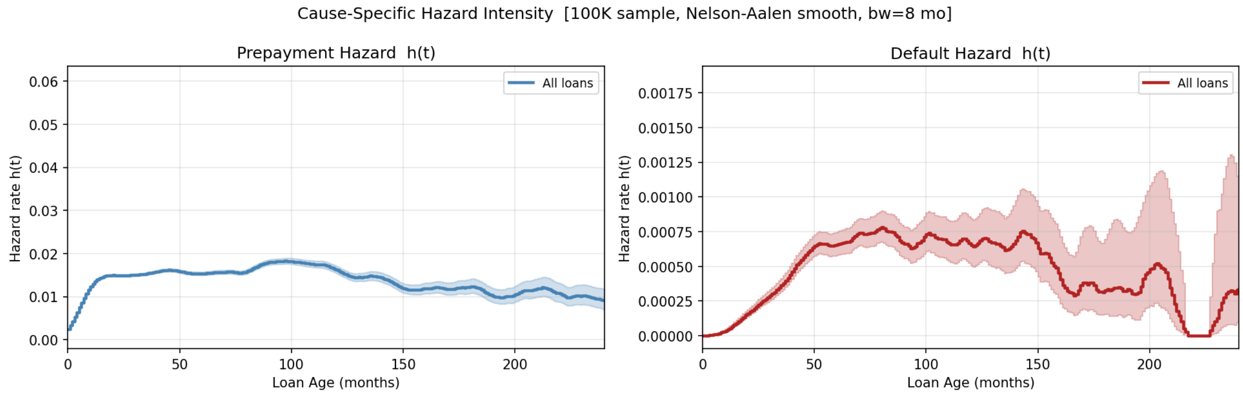

Fig 4. Cause-specific hazard intensity h(t). Prepayment hazard peaks early (months 10–30) then decays — the burnout effect. Default hazard rises slowly and plateaus, reflecting gradual financial deterioration.

Fig 5. Stratified hazard intensity by FICO, LTV, origination rate, and vintage era. High-FICO loans peak earlier (faster refinancing); high-LTV loans show elevated default hazard throughout; GFC vintages (1999–2007) show a default hazard spike absent in post-GFC cohorts.

(b) Aalen-Johansen Cumulative Incidence Functions

The Aalen-Johansen estimator correctly handles competing risks. Treating default as a competing event for prepayment prevents the upward bias that arises from Kaplan-Meier's independence assumption.

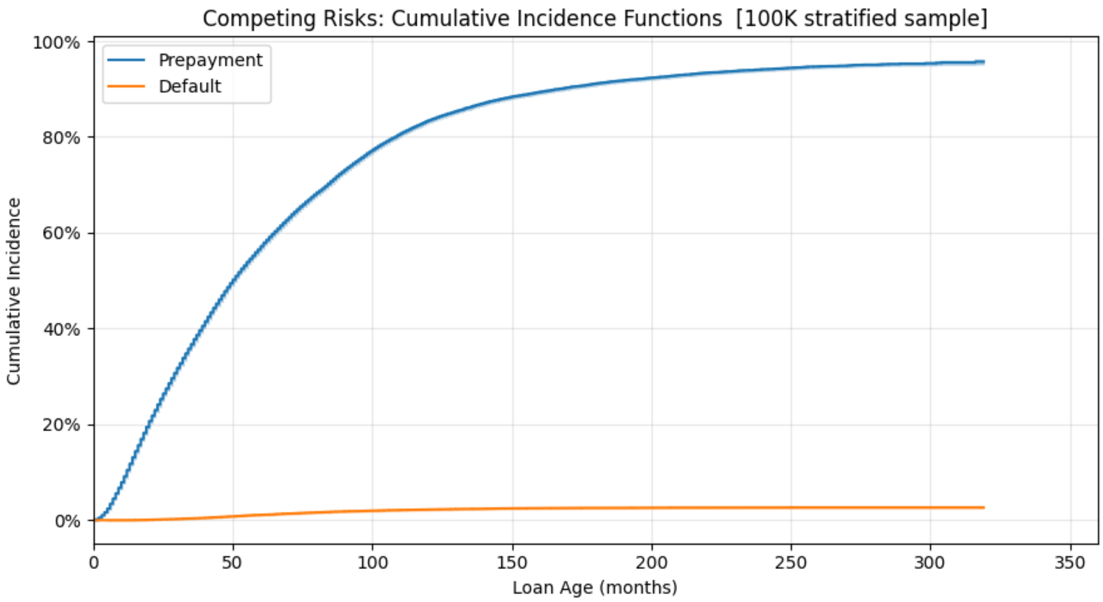

Fig 4. AJ cumulative incidence functions. The 10-year prepayment CIF ≈ 45%; default CIF ≈ 2%.

(c) KM vs AJ — Bias from Ignoring Competing Events

Kaplan-Meier treats default exits as independent censorings — a false assumption since defaulted loans are systematically less likely to prepay. The overestimation grows monotonically with horizon:

- 5-year horizon: KM over-estimates prepayment CIF by ~+0.45 pp
- 10-year horizon: bias ≈ +1.31 pp
- 20-year horizon: bias ≈ +2.23 pp

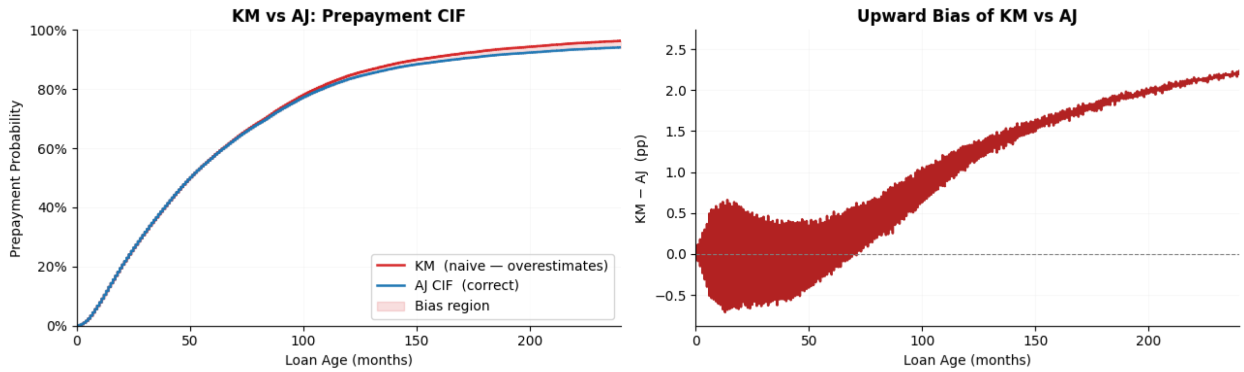

Fig 5. KM vs AJ prepayment CIF and the growing bias.

(d) Stratified CIF by Loan Characteristics

The prepayment/default balance varies substantially across loan segments. High-FICO borrowers show elevated prepayment CIF and near-zero default; high-LTV loans show flatter prepayment curves and elevated default risk; crisis vintages (2006–2008) display default rates an order of magnitude above modern cohorts.

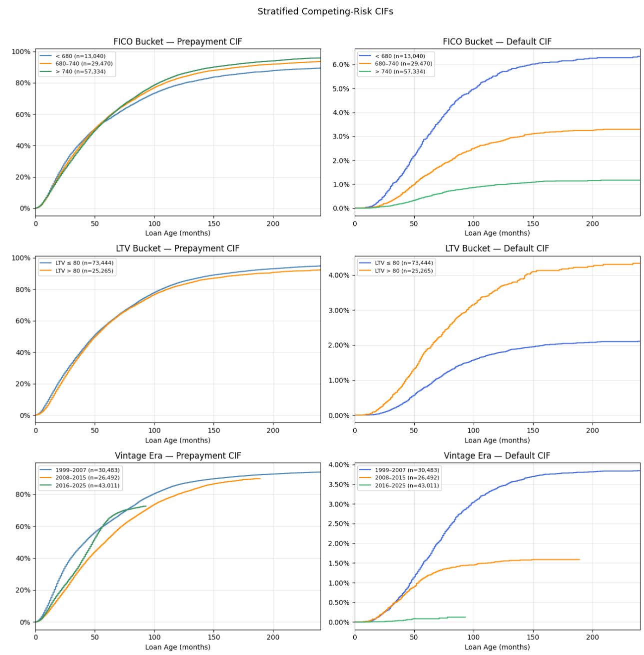

Fig 6. Stratified AJ CIF by FICO bucket, LTV bucket, and vintage era.

(e) Cause-Specific Cox Regression (12 features)

Two separate Cox models are fitted — one for each cause — treating the other cause as censored at exit. Features: 8 numeric (FICO, LTV, DTI, UPB, mortgage rate, unemployment, HPI YoY, rate incentive) + 4 one-hot (loan purpose × 2, occupancy × 2).

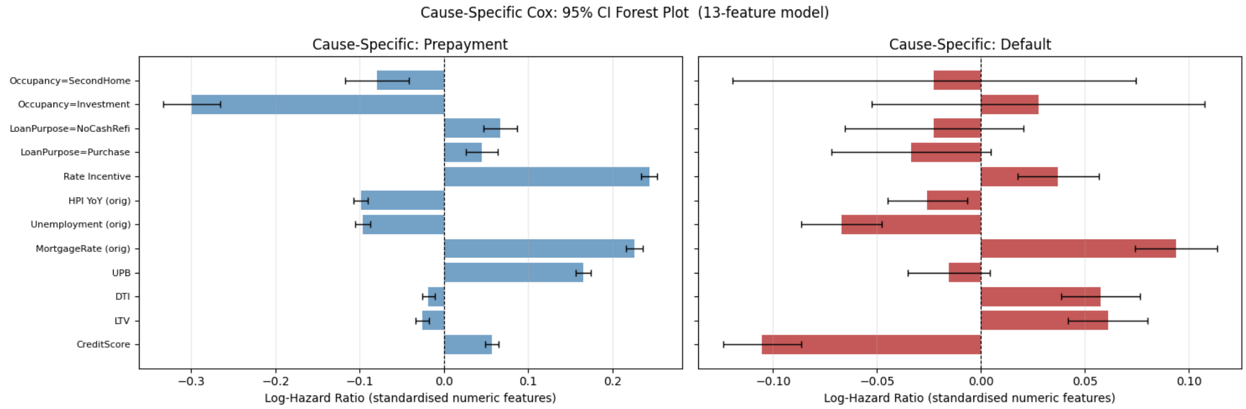

Fig 7. Cause-specific Cox 95% CI forest plot. CreditScore and LTV have opposite signs between causes, illustrating the misrepresentation in a naive combined model.

Key finding: the naive combined-Cox coefficient is a blend of two economically opposite processes. For CreditScore, the prepayment HR is positive (high-FICO borrowers refinance faster) while the default HR is negative (high-FICO borrowers default less). A single coefficient averages these out and misrepresents both risks.

(f) Fine-Gray Subdistribution Hazard

The Fine-Gray model directly targets the CIF rather than the cause-specific hazard. Subjects who experienced a competing event (default) are kept in the risk set with IPCW weights, per Fine & Gray (1999). The subdistribution hazard integrates to the CIF, so covariate effects are directly interpretable in terms of the observable event probability.

Model	Risk set at t	Coefficient interpretation
Cause-specific Cox	All uncensored, non-defaulted	Effect on cause-specific hazard
Fine-Gray	All non-prepaid (incl. defaulted)	**Direct effect on CIF**

For prepayment, **rate_incentive** has a larger coefficient under Fine-Gray than cause-specific Cox: loans with high rate incentive are also less likely to default first, amplifying the competing-risk-adjusted CIF effect.

E(ii) — Time-Dependent Covariates

Standard Cox models fix covariates at origination. The **Andersen-Gill counting-process** extension allows each loan to contribute one row per calendar month with that month's covariate values. The monthly panel dataset (93M rows, 2M loans) provides mortgage rate, unemployment, HPI YoY, and rate incentive updated each period.

Model	Covariate assumption	Data format
Standard Cox — E(i)(e)	Fixed at origination vintage	One row per loan
Andersen-Gill Cox — E(ii)	Updated each month	One row per loan-month

Sample: 10,000 loans stratified by vintage year, yielding ~500K loan-month rows. 11 features: 7 static (FICO, LTV, DTI, UPB, + 3 one-hot) + 4 time-varying (mortgage rate, unemployment, HPI YoY, rate incentive).

Key finding: the current **rate_incentive** coefficient is larger in magnitude under the Andersen-Gill model than the origination-vintage snapshot. A loan originated when rates were high has a permanently large static rate_incentive; the TV model tracks the actual monthly refi option value and is more responsive to rate cycles. Unemployment and HPI also show stronger effects when measured contemporaneously.

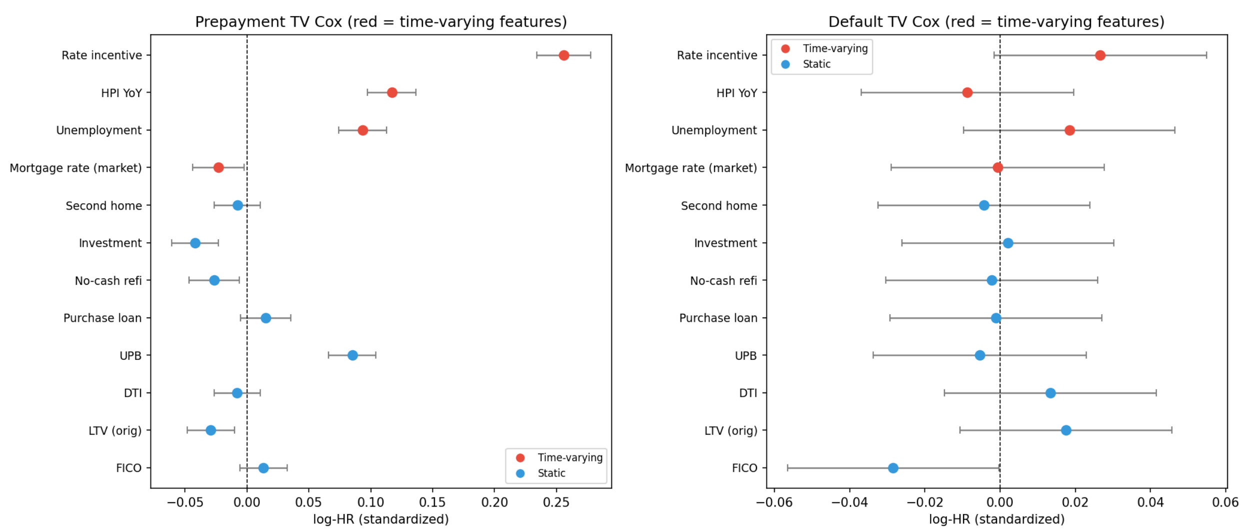

Fig — Eii tv cox forest

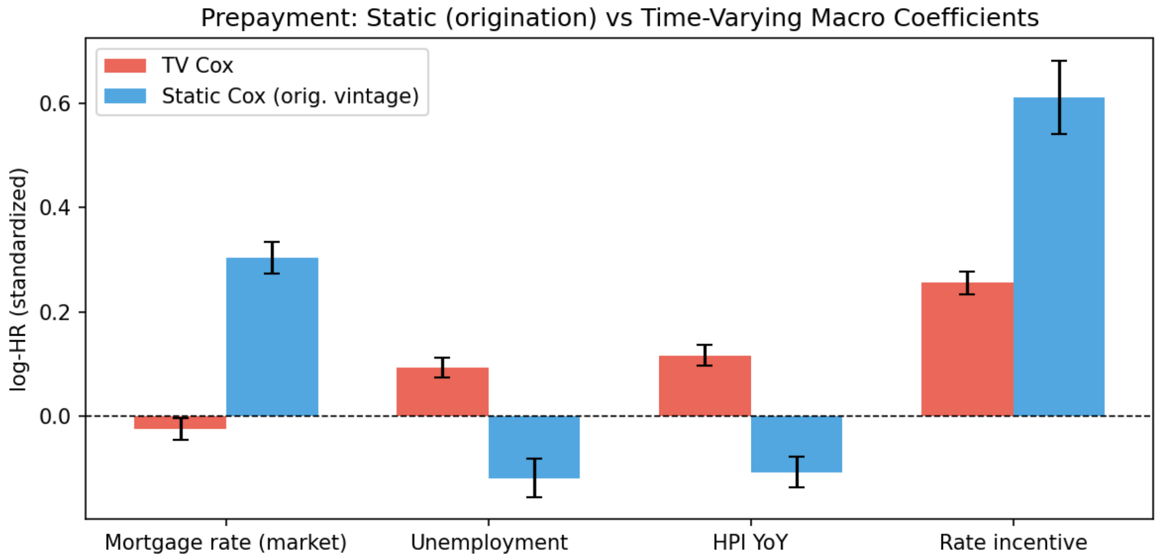

Fig — Eii static vs tv

E(iv) — Scenario Analysis: Interest Rate Shocks

Both Deep Cox and XGBoost are shocked with ±100 bp and ±200 bp changes to *mortgage_rate*. Rate incentive is updated consistently: rate_incentive = orig_rate – (mortgage_rate + shock).

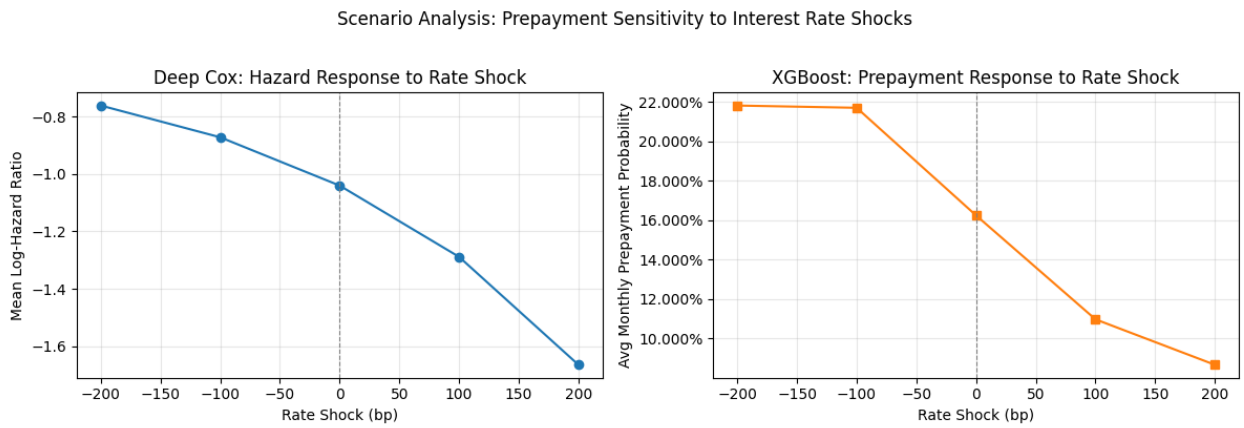

Fig 8. Prepayment sensitivity to interest rate shocks. XGBoost (right) shows clear monotone convex response; Deep Cox (left) shows a smoother log-hazard ratio response.

Interpretation

- XGBoost shows a clear monotone, convex response: rate cuts raise prepayment probability more than equivalent rate rises reduce it.
- Deep Cox responds monotonically with smaller relative changes — the log-hazard is a smoother function of rate_incentive.
- Asymmetry (rate cuts > rate rises) reflects the burnout effect: the most rate-sensitive borrowers have already prepaid, dampening upside response.
- Both models agree on direction, confirming robustness of the rate-incentive channel across architectures.

Summary of Part E Extensions

Section	Method	Key Result
E(i)(b) AJ CIF	Aalen-Johansen	10-yr prepay CIF ≈ 45%; default ≈ 2%
E(i)(c) KM Bias	KM vs AJ comparison	KM over-estimates by +0.5–2.2 pp at 5–20 yr
E(i)(d) Stratified CIF	AJ by FICO / LTV / era	Crisis vintages show 8–12× higher default CIF
E(i)(e) Cause-Specific Cox	Two CoxPH models, 12 features	CreditScore: opposite signs across causes
E(i)(f) Fine-Gray	IPCW-weighted Cox	rate_incentive HR larger vs cause-specific
E(ii) TV Cox	Andersen-Gill, 11 features	Current rate_incentive > origination snapshot
E(iv) Scenarios	Deep Cox + XGBoost shock	Convex, asymmetric prepayment response